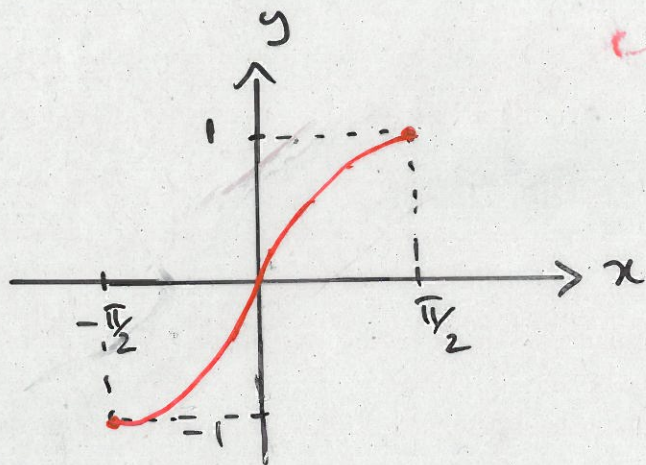


1.10 Inverse trigonometric functions

* let consider

$$y = \sin x, \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$



$$f(x) = \sin x, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

is one-to-one

$$\sin^{-1} x = y \iff \sin y = x \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \\ (-1 \leq x \leq 1)$$

"inverse sine function or arcsine function"

$$\sin^{-1}(\sin x) = x \quad \text{for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\sin(\sin^{-1} x) = x \quad \text{for } -1 \leq x \leq 1$$

Example 1: Evaluate

$$D_{\sin^{-1} x} = [-1, 1] \\ R_{\sin^{-1} x} = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

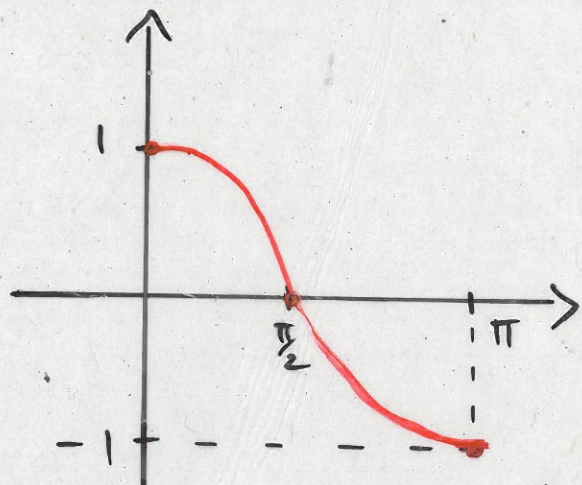
$$(a) \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) ; \cos\left(\sin^{-1}\left(\frac{1}{3}\right)\right)$$

Example 2: solve for x or y ,

$$(a) \sin x = -\frac{1}{2} ; (b) y = \sin^{-1}\left(-\frac{1}{2}\right)$$

* let consider

$$y = \cos x, \quad 0 \leq x \leq \pi$$



$$f(x) = \cos x, \quad x \in [0, \pi]$$

is one-to-one

Its inverse $f^{-1}(x) = \cos^{-1}x$ is called the inverse cosine function or the arccosine function

$$D_{\cos^{-1}x} = [-1, 1], \quad R_{\cos^{-1}x} = [0, \pi]$$

$$\cos^{-1}(\cos x) = x \quad \text{for } 0 \leq x \leq \pi$$

$$\cos(\cos^{-1}x) = x \quad \text{for } -1 \leq x \leq 1$$

Example 3: Evaluate

$$(a) \cos^{-1}\left(-\frac{1}{2}\right) \quad ; \quad (b) \cos^{-1}\left(\cos\left(-\frac{\pi}{2}\right)\right)$$

Example 4: solve for x or y

$$(a) \cos x = 0 \quad ; \quad y = \cos^{-1}0$$

* let consider $y = \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

The inverse tangent function has for domain $\mathbb{R}$ and range $(-\frac{\pi}{2}, \frac{\pi}{2})$

$$\tan^{-1} x = y \iff \tan y = x \text{ and } -\frac{\pi}{2} < y < \frac{\pi}{2}$$

$$\tan^{-1}(\tan x) = x \text{ for } x \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\tan(\tan^{-1} x) = x \text{ for } x \in \mathbb{R}$$

Example 5 : Evaluate

(a) $\tan^{-1}(-\frac{1}{\sqrt{3}})$; (b) ~~sin~~ $(\tan^{-1}(\frac{1}{3}))$

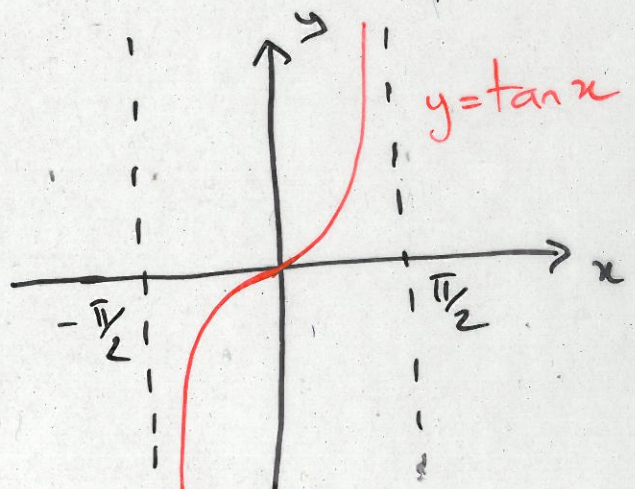
Example 6 : Solve for x

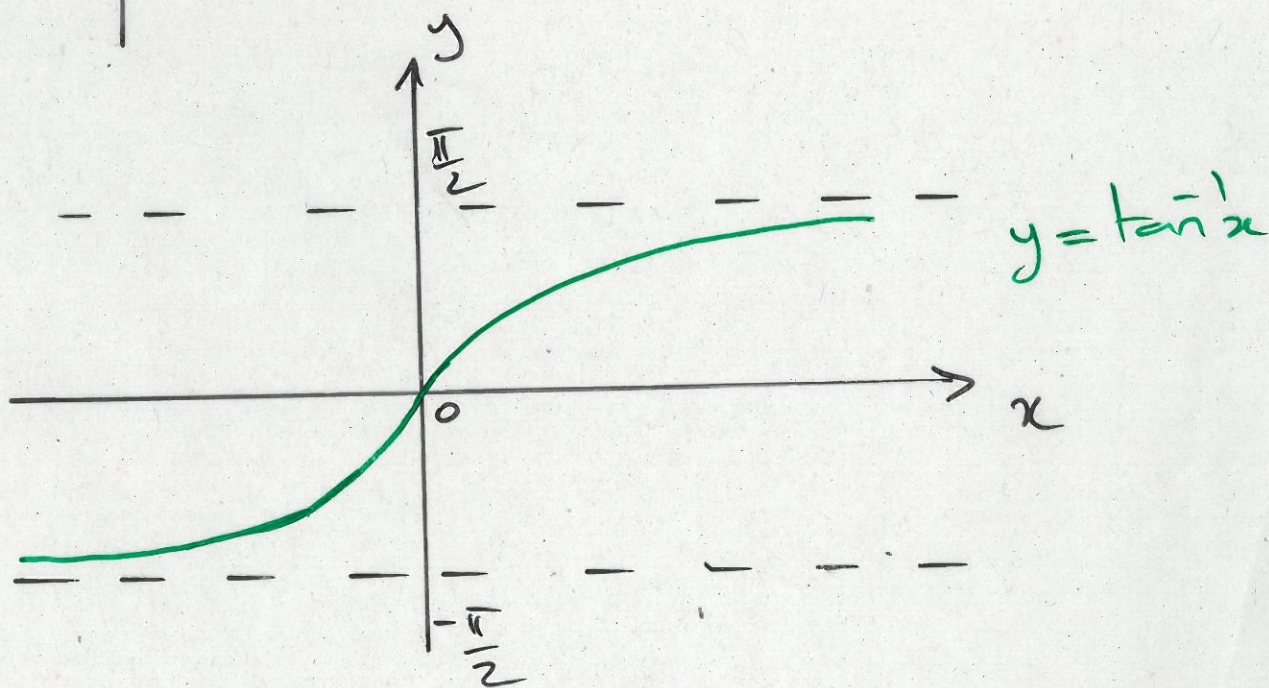
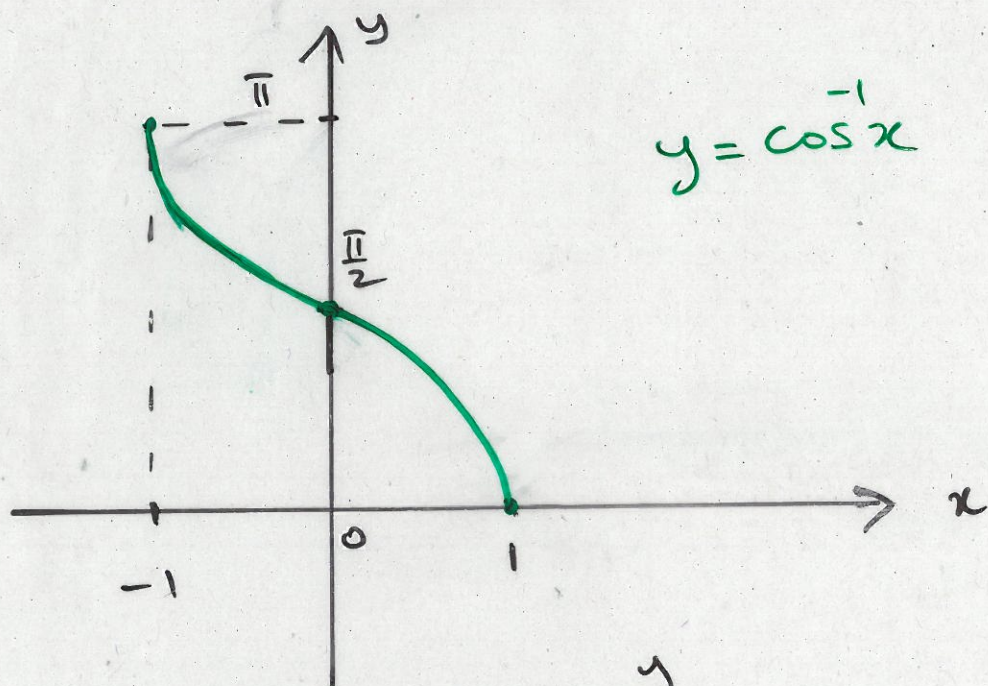
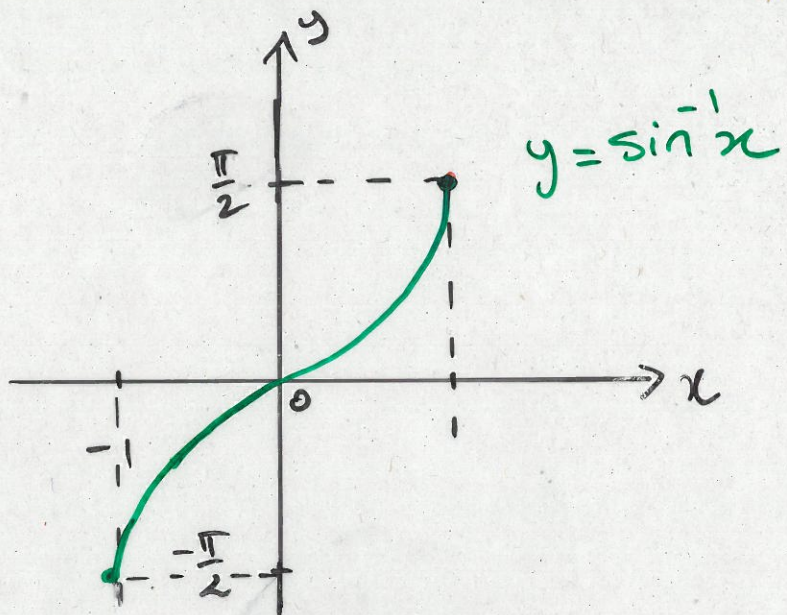
(a) $\tan^{-1}(1) = x$, (b) $\tan x = 1$

Example 7 : Use cancellation properties to simplify

(a) $\sin(\tan^{-1} x)$

(b) $\cos(\sin^{-1} x)$





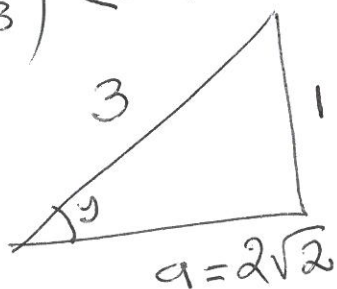
Example 1:

$$(a) x = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \Leftrightarrow \sin x = \frac{\sqrt{3}}{2} \text{ but } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\Rightarrow x = \frac{\pi}{3}$$

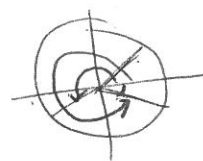
$$(b) x = \cos\left(\sin^{-1}\left(\frac{1}{3}\right)\right)$$

$$y = \sin^{-1}\left(\frac{1}{3}\right) \Leftrightarrow \sin y = \frac{1}{3}, \text{ with } y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



$$\Rightarrow x = \cos(y) = \frac{a}{3} = \frac{2\sqrt{2}}{3}$$

$$\cos\left(\sin^{-1}\left(\frac{1}{3}\right)\right) = \frac{2\sqrt{2}}{3}$$



Example 2:

$$(a) \sin x = -\frac{1}{2} \Rightarrow \text{RA: } \frac{7\pi}{6} \Rightarrow x = \frac{7\pi}{6} + 2k\pi \quad k \in \mathbb{Z}$$

or

$$x = \frac{11\pi}{6} + 2k\pi$$

Infinitely many solutions.

$$(b) y = \sin^{-1}\left(-\frac{1}{2}\right) \Leftrightarrow \sin y = \sin\left(\sin^{-1}\left(-\frac{1}{2}\right)\right) = -\frac{1}{2}$$

and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\Rightarrow y = -\frac{\pi}{6}$$

a unique solution

Example 3:

$$(a) y = \cos^{-1}\left(-\frac{1}{2}\right) \Leftrightarrow \cos(y) = -\frac{1}{2}, y \in [0, \pi]$$

$$\Rightarrow \text{RA: } \frac{\pi}{3}, y = \frac{2\pi}{3}$$

$$(b) x = \cos^{-1}\left(\cos\left(-\frac{\pi}{2}\right)\right) = \cos^{-1}(0) \Rightarrow x = \frac{\pi}{2}$$

(Note that $\cos(-\frac{\pi}{2}) = \cos(\frac{\pi}{2})$ and $-\frac{\pi}{2} \notin [0, \pi]$)

Example 4:

$$(a) \cos x = 0 \Rightarrow x = \frac{\pi}{2} + 2k\pi \text{ or } x = \frac{3\pi}{2} + 2k\pi$$

$$k \in \mathbb{Z}$$

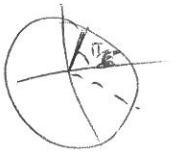
$$(b) y = \cos^{-1}(0) \Rightarrow y = \frac{\pi}{2}$$

Example 5

$$(a) y = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) \Leftrightarrow \tan(y) = -\frac{1}{\sqrt{3}}, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

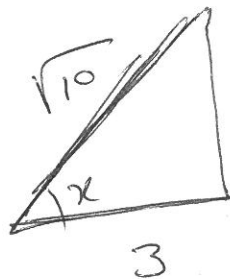
$$\Rightarrow \text{R.A.} = -\frac{\pi}{6}$$

$$y = -\frac{\pi}{6}$$



$$(b) y = \sin\left(\tan^{-1}\left(\frac{1}{3}\right)\right)$$

$$\tan^{-1}\left(\frac{1}{3}\right) = x \Leftrightarrow \tan x = \frac{1}{3}, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$



$$\Rightarrow y = \sin(x) = \frac{1}{\sqrt{10}}$$

Example 6:

$$a) \tan^{-1}(1) = x \Leftrightarrow \tan(\tan^{-1}x) = \tan x$$

$$\Leftrightarrow \tan x = 1, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow x = \frac{\pi}{4}$$

$$(b) \tan x = 1 \Rightarrow x = \frac{\pi}{4} + 2k\pi, \quad k \in \mathbb{Z}.$$

$$\text{or} \\ x = \frac{5\pi}{4} + 2k\pi, \quad k \in \mathbb{Z}.$$

Example 7

let $y = \tan^{-1}(x)$, then $\tan y = x$, $-\frac{\pi}{2} < y < \frac{\pi}{2}$

$$\sec^2 y = 1 + \tan^2 y = 1 + x^2$$

$$\Rightarrow \sec^2 y = \sqrt{1 + x^2} \quad \text{since } \sec y > 0 \text{ for } y \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\Rightarrow \sin(\tan^{-1} x) = \sin y = \cos y \cdot \tan y$$

$$= \frac{1}{\sec y} \cdot \tan y$$

$$= \frac{1}{\sqrt{1+x^2}} \cdot x$$

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}$$

(b) let $y = \sin^{-1} x$, then $\sin y = x$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$$\Rightarrow \cos^2 y = 1 - \sin^2 y = 1 - x^2$$

$$\Rightarrow \cos y = \sqrt{1-x^2} \quad \text{since } \cos y \geq 0 \text{ for } y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\Rightarrow \cos(\sin^{-1} x) = \sqrt{1-x^2}$$